

Deep Learning - Theory and Practice

IE 643

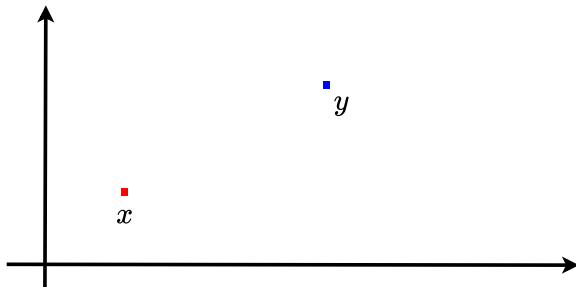
Lecture 11 (Part 2)

September 4, 2025

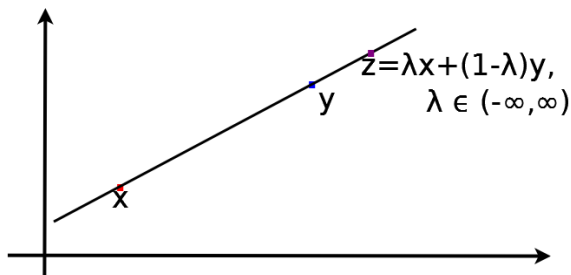
- 1 Convex Sets
- 2 Convex Functions in 1D
- 3 Convex Sets and Functions in Higher Dimensions
- 4 Strictly Convex Functions
- 5 Convex Optimization

Convex Sets

Points in a 2D space

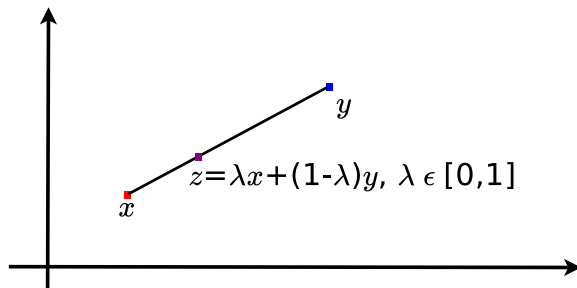


Affine combination of two points



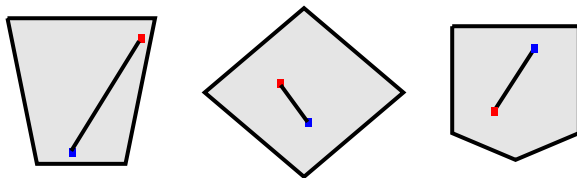
- z is an arbitrary point on the line passing through x and y .

Convex combination of two points



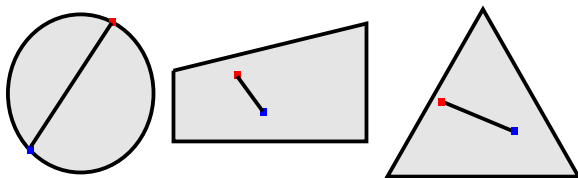
- z is an arbitrary point on the line segment connecting x and y .

Convex Sets



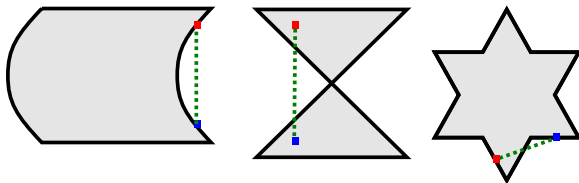
- A set $\mathcal{C}$ is convex if $\lambda x + (1 - \lambda)y \in \mathcal{C}$, $\forall x, y \in \mathcal{C}$, $\forall \lambda \in [0, 1]$.
- The line segment connecting x and y in $\mathcal{C}$ lies entirely within $\mathcal{C}$.

Convex Sets



- A set $\mathcal{C}$ is convex if $\lambda x + (1 - \lambda)y \in \mathcal{C}$, $\forall x, y \in \mathcal{C}$, $\lambda \in [0, 1]$.
- The line segment connecting x and y in $\mathcal{C}$ lies entirely within $\mathcal{C}$.

Non-convex Sets



- A set $\mathcal{C}$ is not convex if there exist two points x and y in $\mathcal{C}$ such that $\lambda x + (1 - \lambda)y \notin \mathcal{C}$, for some $\lambda \in [0, 1]$.
- The line segment connecting x and y in $\mathcal{C}$ does not entirely lie within $\mathcal{C}$.

Convex Sets and Convex Combination

Going beyond two points

- Let x, y, z be points in a set $\mathcal{C}$.
- How to extend the definition of convex combination to these three points?

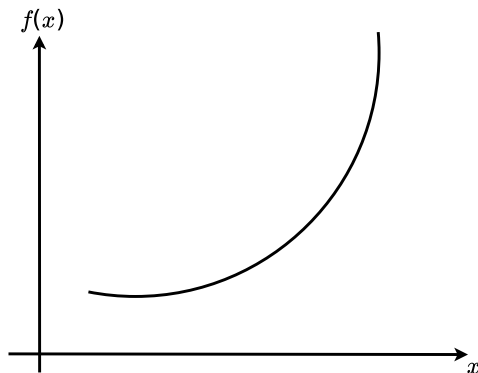
Convex Sets and Convex Combination

Going beyond two points

- Let x, y, z be points in a set $\mathcal{C}$.
- How to extend the definition of convex combination to these three points? (Homework!)

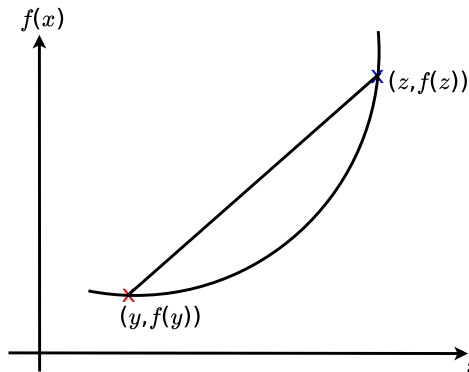
Convex Functions

Convex Function - Definition



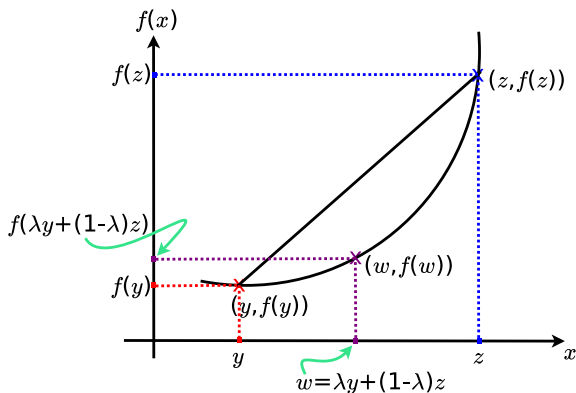
- A function $f : \mathcal{C} \rightarrow \mathbb{R}$, defined over a **convex set** $\mathcal{C} \subseteq \mathbb{R}$ is called convex if $f(\lambda y + (1 - \lambda)z) \leq \lambda f(y) + (1 - \lambda)f(z)$, $\forall y, z \in \mathcal{C}$, $\forall \lambda \in [0, 1]$.

Convex Function - Definition



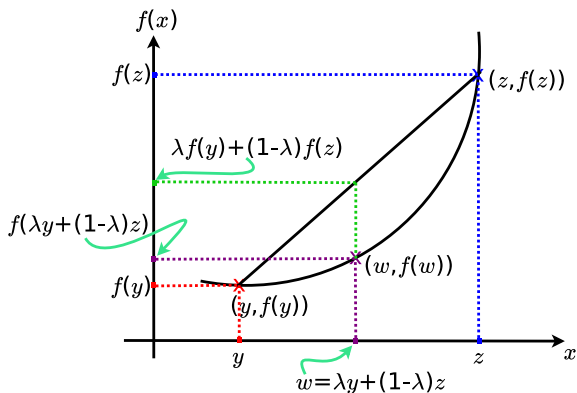
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- Chord over-estimates the graph of function.

Convex Function - Definition



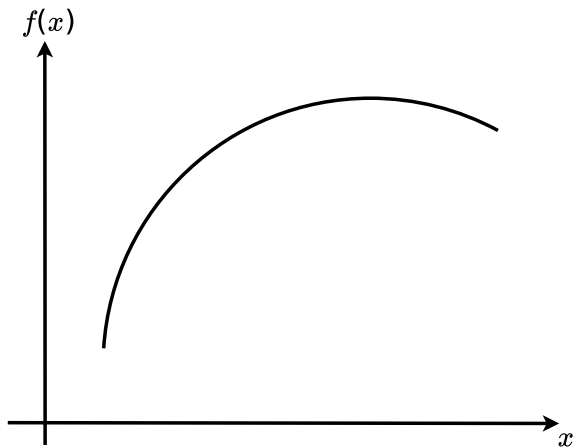
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Convex Function - Definition



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- Chord over-estimates the graph of function.

Concave Function



- Concave function is a close relative of convex function.
- A function $f : \mathcal{C} \rightarrow \mathbb{R}$, defined over a **convex set** $\mathcal{C} \subseteq \mathbb{R}$ is called concave if $-f$ is convex over $\mathcal{C}$.

Note: Concave functions are also defined over convex sets.

Convex Function - Characterization

Extending convex function definition to more than two points.

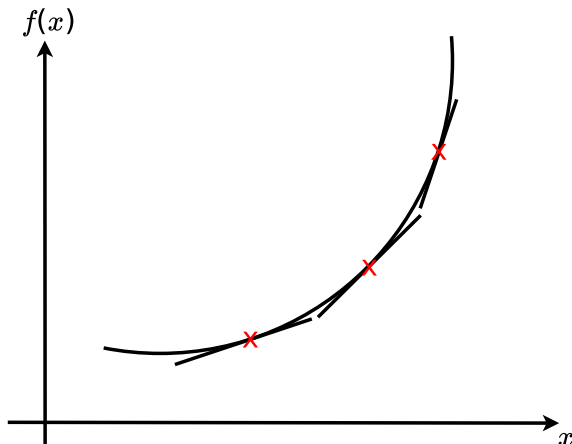
How to extend the definition of convex functions to a set of points
 $\{x^1, x^2, \dots, x^m\} \subset \mathcal{C}$?

Convex Function - Characterization

Extending convex function definition to more than two points.

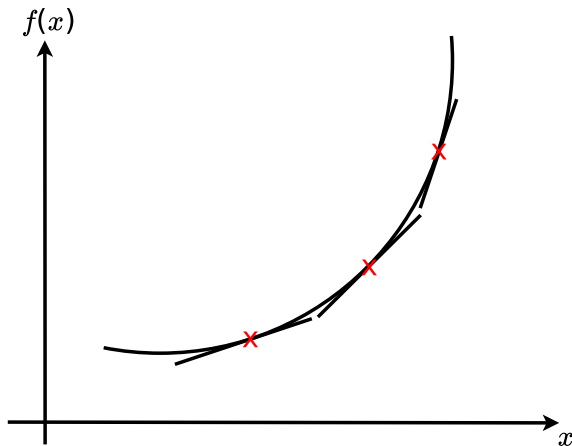
How to extend the definition of convex functions to a set of points $\{x^1, x^2, \dots, x^m\} \subset \mathcal{C}$? (Homework !)

Convex Function - First Order Characterization



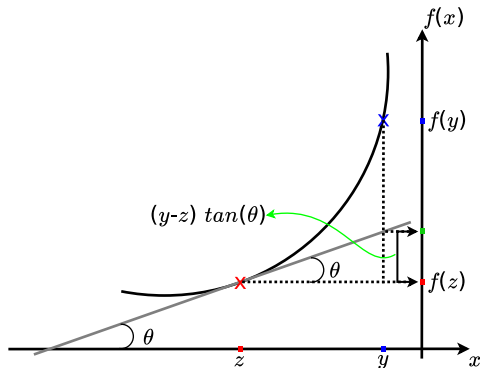
- For **differentiable** convex functions, the tangent lines under-estimate the graph of function.
- Recall: Tangent at a point is a **first-order approximation** of a function.

Convex Function - First Order Characterization



- Recall: **First order approximation** of a function f at y in the vicinity of point z :
 - $f(y) \approx f(z) + (y - z)f'(z)$.

Convex Function - First Order Characterization



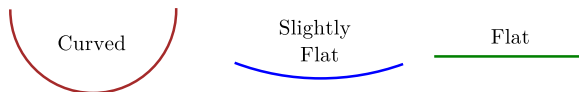
- Let $\mathcal{C} \subseteq \mathbb{R}$ be an **open interval**. Let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a **continuously differentiable** function. Then f is convex if and only if

$$f(y) \geq f(z) + (y - z)f'(z), \quad \forall y, z \in \mathcal{C}.$$

- $f'(z)$ is the derivative of f at z .

Note: $\mathcal{C} \subseteq \mathbb{R}$ is assumed to be an **open interval**.

Convex Function - Second Order Characterization



- Let $\mathcal{C} \subseteq \mathbb{R}$ be an open interval. Let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a **twice continuously differentiable** function. Then f is convex if and only if $f''(x) \geq 0, \forall x \in \mathcal{C}$.
- $f''(x)$ is the double derivative of f at x .
- $f''(x) \geq 0$ indicates **non-negative curvature**.

Convex Function - Interesting Properties

Convex functions enjoy several interesting properties.

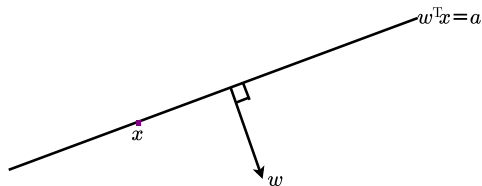
- If f_i are convex for $i = 1, \dots, m$, their **weighted sum** $\sum_{i=1}^m \theta_i f_i$ is convex, when $\theta_i \geq 0, \forall i = 1, \dots, m$.
- If f_i are convex for $i = 1, \dots, m$, **$\max_{i=1, \dots, m} f_i$** is convex.
- If f is convex then $g(x) = f(ax + b), a, b \in \mathbb{R}$ is also convex. (**Affine invariance**)

Moving Towards Higher Dimensions...

Convex Sets In High Dimensions

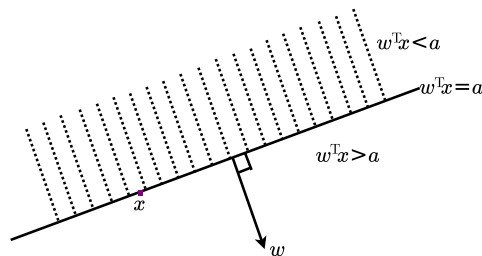
- Extreme cases: the empty set $\emptyset$ and the full space $\mathbb{R}^d$.

Convex Sets In High Dimensions



- Hyperplane: $\{x \in \mathbb{R}^d : w^T x = a\}$, for some $\mathbf{0} \neq w \in \mathbb{R}^d$ and $a \in \mathbb{R}$.

Convex Sets In High Dimensions



- Closed Halfspace:

- ▶ $\{x \in \mathbb{R}^d : w^T x \geq a\}$, for some $0 \neq w \in \mathbb{R}^d$ and $a \in \mathbb{R}$.
- ▶ $\{x \in \mathbb{R}^d : w^T x \leq a\}$, for some $0 \neq w \in \mathbb{R}^d$ and $a \in \mathbb{R}$.

- Open Halfspace:

- ▶ $\{x \in \mathbb{R}^d : w^T x > a\}$, for some $0 \neq w \in \mathbb{R}^d$ and $a \in \mathbb{R}$.
- ▶ $\{x \in \mathbb{R}^d : w^T x < a\}$, for some $0 \neq w \in \mathbb{R}^d$ and $a \in \mathbb{R}$.

High Dimensional Representation - Notations

- Gradient of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ at a point x

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \vdots \\ \frac{\partial f(x)}{\partial x_d} \end{pmatrix}$$

High Dimensional Representation - Notations

- Hessian Matrix of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ at a point x

$$H = \nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_d} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_d} \\ \vdots & \vdots & \cdots & \vdots \\ \vdots & \vdots & \cdots & \vdots \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_d \partial x_1} & \frac{\partial^2 f(x)}{\partial x_d \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_d \partial x_d} \end{pmatrix}$$

- Note the size of H : $d \times d$, we will denote this as $H \in \mathbb{R}^{d \times d}$.
- Note also that H is **symmetric**. (why?)

Transpose Of A Matrix

Matrix A of size $d \times d$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1d} \\ a_{21} & a_{22} & \dots & a_{2d} \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ a_{d1} & a_{d2} & \dots & a_{dd} \end{pmatrix}$$

Transpose of A (of same size)

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{d1} \\ a_{12} & a_{22} & \dots & a_{d2} \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ a_{1d} & a_{2d} & \dots & a_{dd} \end{pmatrix}$$

Note: Rows of matrix A are columns of matrix A^T .

Symmetric Matrix

- A matrix A is symmetric if $A = A^\top$.

$$A = \begin{pmatrix} a_{11} & \textcolor{red}{a}_{12} & \dots & \textcolor{green}{a}_{1d} \\ \textcolor{red}{a}_{12} & a_{22} & \dots & \textcolor{blue}{a}_{2d} \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ \textcolor{green}{a}_{1d} & \textcolor{blue}{a}_{2d} & \dots & a_{dd} \end{pmatrix} = A^\top$$

Symmetric Positive (Semi-)definite Matrix

- A symmetric matrix $A \in \mathbb{R}^{d \times d}$ is called **positive semi-definite** (denoted by $A \succeq 0$) if

$$x^T A x \geq 0, \forall x \in \mathbb{R}^d.$$

Caution: This definition is non-intuitive.

- A symmetric matrix $A \in \mathbb{R}^{d \times d}$ is called **positive definite** (denoted by $A \succ 0$) if

$$x^T A x > 0, \forall x \in \mathbb{R}^d \text{ such that } x \neq 0.$$

Symmetric Positive (Semi-)definite Matrix

Computation-friendly definitions

- $A \succeq 0 \iff$ all eigen values of A are non-negative.
- $A \succ 0 \iff$ all eigen values of A are positive.
- **Recall:**
 - ▶ A (non-zero) vector x is called an eigen vector of matrix $A \in \mathbb{R}^{d \times d}$ with a corresponding eigen value β , if $Ax = \beta x$.
 - ▶ An eigen value of a symmetric matrix is always real. (Why?)

Convex Function - Characterization In 1D

Let $\mathcal{C} \subseteq \mathbb{R}$ be an open convex set and let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a convex function.

Recall:

- Zero-th Order Characterization

$$f(\lambda y + (1 - \lambda)z) \leq \lambda f(y) + (1 - \lambda)f(z), \quad \forall y, z \in \mathcal{C}, \quad \forall \lambda \in [0, 1].$$

- First Order Characterization

$$f \text{ continuously differentiable, } f(y) \geq f(z) + f'(z)(y - z), \quad \forall y, z \in \mathcal{C}.$$

- Second Order Characterization

$$f \text{ twice continuously differentiable and } f''(x) \geq 0, \quad \forall x \in \mathcal{C}.$$

Convex Function - Characterization In High Dimensions

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be an open convex set and let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a convex function.

- Zero-th Order Characterization

$$f(\lambda y + (1 - \lambda)z) \leq \lambda f(y) + (1 - \lambda)f(z), \quad \forall y, z \in \mathcal{C}, \quad \forall \lambda \in [0, 1].$$

- First Order Characterization

$$f \text{ continuously differentiable, } f(y) \geq f(z) + \nabla f(z)^\top (y - z), \quad \forall y, z \in \mathcal{C}.$$

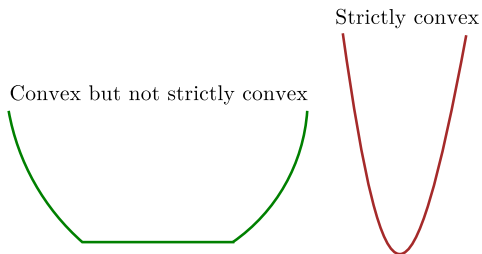
- Second Order Characterization

$$f \text{ twice continuously differentiable and } \nabla^2 f \succeq 0.$$

Other Flavors Of Convex Function

- Strictly convex function
- Strongly convex function

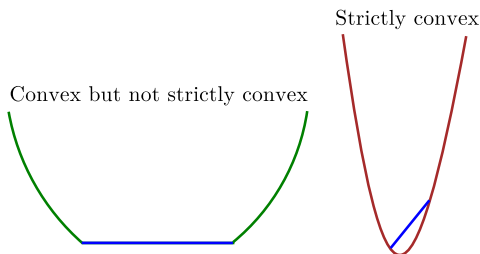
Strictly Convex Function



A function $f : \mathcal{C} \rightarrow \mathbb{R}$ defined over a convex set $\mathcal{C}$ is called **strictly convex** if

$$f(\lambda y + (1 - \lambda)z) < \lambda f(y) + (1 - \lambda)f(z), \quad \forall y, z \in \mathcal{C} \text{ s.t. } x \neq y, \quad \forall \lambda \in (0, 1).$$

Strictly Convex Function

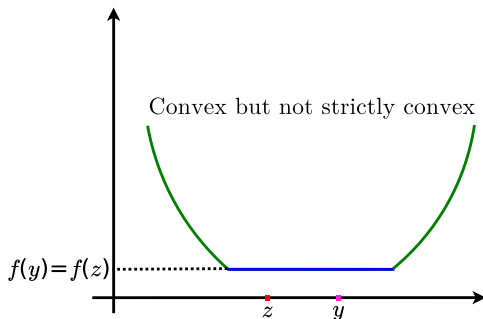


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- Graph of the function should be **strictly below** the chord !

Strictly Convex Function - First Order Characterization



- Let $\mathcal{C} \subseteq \mathbb{R}$ be an open convex set. Let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a **continuously differentiable** function. Then f is **strictly convex** if and only if

$$f(y) > f(z) + \nabla f(z)^\top (y - z), \forall y, z \in \mathcal{C}, y \neq z.$$

Strictly Convex Function - Second Order Characterization

- Let $\mathcal{C} \subseteq \mathbb{R}$ be an open convex set. Let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a **twice continuously differentiable** function. f is **strictly convex** if

$$\nabla^2 f \succ 0.$$

- Important note:** This positive definiteness condition is sufficient but not necessary.
 - e.g. $f(x) = x^4$, $x \in \mathbb{R}$ is strictly convex but $f''(x) = 0$ at $x = 0$.

Strictly Convex Function

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be a convex set and let $f : \mathcal{C} \rightarrow \mathbb{R}$ be a **strictly convex** function.

- **Zero-th Order Characterization**

$$f(\lambda y + (1 - \lambda)z) < \lambda f(y) + (1 - \lambda)f(z), \quad \forall y, z \in \mathcal{C}, \quad y \neq z \quad \forall \lambda \in (0, 1).$$

- **First Order Characterization**

f continuously differentiable in $\text{int}(\mathcal{C})$ and

$$\forall z \in \text{int}(\mathcal{C}), \quad f(y) > f(z) + \nabla f(z)^\top (y - z), \quad \forall y \neq z \in \mathcal{C}.$$

Strictly Convex Function

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be an open convex set and let $f : \mathcal{C} \rightarrow \mathbb{R}$.

- **Second Order Characterization (sufficient condition)**

f twice continuously differentiable and $\nabla^2 f \succ 0$, $\implies$

f is strictly convex.

Convex Optimization Problem

General Optimization Problem

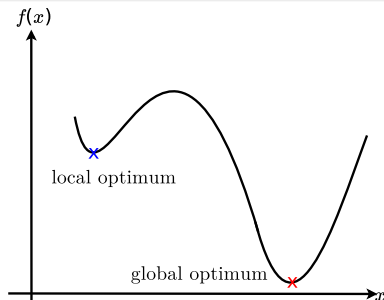
$$\min_{x \in \mathcal{C}} f(x)$$

- f is called **objective function** and $\mathcal{C}$ is called **feasible set**.
- Let $f^* = \min_{x \in \mathcal{C}} f(x)$ denote the **optimal objective function value**.
- **Optimal Solution Set** $X^* = \{x \in \mathcal{C} : f(x) = f^*\}$.
- Let us denote by x^* an optimal solution in X^* .

General Optimization Problem

$$\min_{x \in \mathcal{C}} f(x)$$

(OP)



General Optimization Problem

$$\min_{x \in \mathcal{C}} f(x) \quad (\text{OP})$$

Local Optimal Solution

A solution z to (OP) is called local optimal solution if $f(z) \leq f(\hat{z})$, $\forall \hat{z} \in \mathcal{N}_\epsilon(z)$ for some $\epsilon > 0$.

Recall: $\mathcal{N}_\epsilon(z)$ denotes the ϵ -neighborhood of z with respect to a suitable distance metric.

Global Optimal Solution

A solution z to (OP) is called global optimal solution if $f(z) \leq f(\hat{z})$, $\forall \hat{z} \in \mathcal{C}$.

Local vs. Global Optimal Solutions

First-order necessary condition for optimality

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a **continuously differentiable** function. If a point $x^* \in \mathbb{R}^d$ is a local optimal solution of $\min_{x \in \mathbb{R}^d} f(x)$ then $\nabla f(x^*) = \mathbf{0}$.

General Optimization Problem

$$\min_{x \in \mathcal{C}} f(x)$$

- $\mathcal{C} \subseteq \mathbb{R}^d$ is a convex set
- $f : \mathcal{C} \rightarrow \mathbb{R}$ is a convex function

Convex Optimization Regime!

What Is So Special About Convex Optimization?

- Appealing geometry in small dimensions
- Nice properties from an optimization perspective
 - ▶ Every local optimal solution (if it exists) is a global optimal solution.

Local vs. Global Optimal Solutions

Proposition

Consider the convex optimization problem $\min_{x \in \mathbb{R}^d} f(x)$ and let $X^* \neq \emptyset$ (**recall**: $X^* = \{x \in \mathbb{R}^d : f(x) \leq f(z) \forall z \in \mathbb{R}^d\}$ denotes the set of optimal solutions of the optimization problem).

Then every local optimal solution of the problem $\min_{x \in \mathbb{R}^d} f(x)$ is a global optimal solution.

Local vs. Global Optimal Solutions

First-order necessary and sufficient condition for optimality

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a **continuously differentiable** convex function. A point $x^* \in \mathbb{R}^d$ is an optimal solution of $\min_{x \in \mathbb{R}^d} f(x)$ **if and only if** $\nabla f(x^*) = \mathbf{0}$.

Local vs. Global Optimal Solutions

Proposition

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a **continuously differentiable** convex function. A point $x^* \in \mathbb{R}^d$ is an optimal solution of $\min_{x \in \mathbb{R}^d} f(x)$ **if and only if** $\nabla f(x^*) = \mathbf{0}$.

Note: The zero gradient condition is necessary and sufficient for optimality!

Optimal solutions for Strictly Convex Functions

Uniqueness of solution

Consider the convex optimization problem $\min_{x \in \mathbb{R}^d} f(x)$ where f is **strictly convex**. If the set $X^* = \operatorname{argmin}_{x \in \mathbb{R}^d} f(x)$ is non-empty, then X^* contains **exactly one** element.